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Modeling and Simulation of Quadrotor UAV Using PID Controller

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***Abstract*—Recent developments in sensor technology, actuators and energy storage devices provides opportunities for development of micro flying robots. This paper provides information about modeling and control of a Quadrotor. Proportional Integral Derivative (PID) controller is designed for altitude and attitude control of Quadrotor. Various simulations were performed and it could be interpreted that the PID controller is capable of tracking the desired reference values. Finally, the last section is devoted for discussing the results of each simulation.**

Keywords—quadrotor, UAV control, quadcopter, dynamic modeling, PID controller, mathematical model, attitude control, altitude control

I. INTRODUCTION

The wide range of uses in both defence industry and civilian markets was supporting the funding of UAV based projects. Search and rescue, cargo carriage, forest fire detection, crop dusting, defense industry and security applications are major fields that UAVs could be used. [1], [2]

In this paper, the nonlinear mathematical model of the quadrotor is derived. Based on the nonlinear mathematical model, a quadrotor simulator is developed in Matlab Simulink for an OS4 micro quadrotor platform. Finally, PID controller approach is presented and its performance is demonstrated.

II. QUADROTOR CONFIGURATION

Quadrotors have four rotors that they could be controlled separately. This makes possible to move in six degrees of freedom (6 DOF). Six degrees of freedom consists of three translational and three rotational DOF. Quadrotors have 4 propellers and these propellers could be configured in cross or plus configuration. In this paper, cross configuration is used. Quadrotor's (1,3) and (2,4) pairs of propellers counterly rotate in each other. Changing the rotor speed of quadrotor produces lift force and generates movement. In this way, increasing or reducing all of the 4 propeller's speeds jointly changes vertical movement. By varying the 2 and 4 propeller's speed inversely generates roll rotation combined with lateral movement. Changing 1 and 3 propeller's speed conversely produces pitch rotation and the corresponding lateral motion. Yaw rotation is more different than others. It arises from the difference in the counter-torque between each pair of propellers. Fig. 1 demonstrates the concept movement of quadrotor. [3], [4], [5]

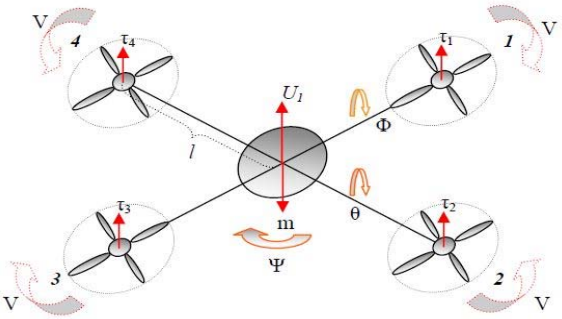


Fig. 1. Quadrotors' movements and angles description

III. QUADROTOR DYNAMIC MODELLING

There are many studies about derivation and analysis of the mathematical model of quadcopters in the scientific literature. The nonlinear mathematical model of a quadcopter is summarized in this section.

A. The Mathematical Model of Quadrotor

Conversion from earth frame to body frame is possible. Equation (1) through "(5)" demonstrate the transition between earth and body frames. (cos : c and sin : s) [6], [7].

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \mathrm{R} * \begin{bmatrix} u \\ v \\ w \end{bmatrix} \tag{1}$$

$$\mathrm{R}(\psi) = \begin{bmatrix} \cos(\psi) & \sin(\psi) & 0 \\ -\sin(\psi) & \cos(\psi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2}$$

$$\mathrm{R}(\theta) = \begin{bmatrix} \cos(\theta) & 0 & -\sin(\theta) \\ 0 & 1 & 0 \\ \sin(\theta) & 0 & \cos(\theta) \end{bmatrix} \tag{3}$$

$$\mathrm{R}(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\varphi) & \sin(\varphi) \\ 0 & -\sin(\varphi) & \cos(\varphi) \end{bmatrix} \tag{4}$$

$$\mathrm{R}(\varphi, \theta, \psi) = \mathrm{R}(\varphi)\,\mathrm{R}(\theta)\,\mathrm{R}(\psi) \tag{5}$$

R is the rotation matrix from body to inertial frame and is given with eqn. 6.

$$\mathrm{R} = \begin{bmatrix} c\theta c\psi & c\theta s\psi & -s\theta \\ s\theta s\varphi c\psi - s\psi c\varphi & s\psi s\theta s\varphi + c\psi c\varphi & s\varphi c\theta \\ s\theta c\varphi c\psi + s\psi s\varphi & s\psi s\theta c\varphi - c\psi s\varphi & c\varphi c\theta \end{bmatrix} \tag{6}$$

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The following equations (7) and (8) reflects the relation between the Euler angle rates and the body angular rates.

$$\begin{bmatrix} P \\ Q \\ R \end{bmatrix} = \begin{bmatrix} 1 & 0 & -\sin(\theta) \\ 0 & \cos(\varphi) & \sin(\varphi)\cos(\theta) \\ 0 & -\sin(\varphi) & \cos(\varphi)\cos(\theta) \end{bmatrix} \begin{bmatrix} \varphi' \\ \theta' \\ \psi' \end{bmatrix} \quad (7)$$

$$\begin{bmatrix} \Phi' \\ \theta' \\ \psi' \end{bmatrix} = \begin{bmatrix} 1 & \sin(\varphi)\tan(\theta) & \cos(\varphi)\tan(\theta) \\ 0 & \cos(\varphi) & -\sin(\varphi) \\ 0 & \sin(\varphi)\sec(\theta) & \cos(\varphi)\sec(\theta) \end{bmatrix} \begin{bmatrix} P \\ Q \\ R \end{bmatrix} \quad (8)$$

Fi and Ti are the thrust and torque produced by each rotor (i = 1; 2; 3; 4). wi is ith motor speed, b is a positive thrust factor, d is drag factor, $w_r = -w_1 + w_2 - w_3 + w_4$ and lastly l is the arm length of the quadrotor. [8]

$$Fi = bwi^2 \quad (9)$$

$$Ti = dwi^2 \quad (10)$$

Using Newton-Euler equations, the force and torque equations could be defined in ‘‘(11)”. [9]

$$u = \begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ U_4 \end{bmatrix} = \begin{bmatrix} F \\ T_\varphi \\ T_\theta \\ T_\psi \end{bmatrix} = \begin{bmatrix} b & b & b & b \\ 0 & -lb & 0 & lb \\ lb & 0 & -lb & 0 \\ -d & d & -d & d \end{bmatrix} \begin{bmatrix} w_1^2 \\ w_2^2 \\ w_3^2 \\ w_4^2 \end{bmatrix} \quad (11)$$

U_1 is the total lift force; U_2, U_3, U_4 are the torques acting on quadrotor. Omega to force & torques’ block has following equations.

$$\begin{bmatrix} w_1^2 \\ w_2^2 \\ w_3^2 \\ w_4^2 \end{bmatrix} = \begin{bmatrix} \frac{1}{4b} & 0 & \frac{1}{2bl} & \frac{1}{4d} \\ \frac{1}{4b} & -\frac{1}{2bl} & 0 & \frac{1}{4d} \\ -\frac{1}{4b} & 0 & -\frac{1}{2bl} & -\frac{1}{4d} \\ \frac{1}{4b} & \frac{1}{2bl} & 0 & \frac{1}{4d} \end{bmatrix} \begin{bmatrix} U_1 \\ U_2 \\ U_3 \\ U_4 \end{bmatrix} \quad (12)$$

The dynamic model of the quadrotor with the x, y, z motions as a result of translational and rotational movement is in equation (13) through “(18)".

$$\ddot{X} = (\sin\theta\cos\varphi\cos\psi + \sin\psi\sin\varphi)\frac{U1}{m} \quad (13)$$

$$\ddot{Y} = (\sin\psi\sin\theta\cos\varphi - \cos\psi\sin\varphi)\frac{U1}{m} \quad (14)$$

$$\ddot{Z} = (\cos\varphi\cos\theta)\frac{U1}{m} - g \quad (15)$$

$$\ddot{\varphi} = \left(\frac{I_y - I_z}{I_x}\right)\dot{\theta}\dot{\psi} + \frac{J_R}{I_x}\dot{\theta}w_r + \frac{l}{I_x}U_2 \quad (16)$$

$$\ddot{\theta} = \left(\frac{I_z - I_x}{I_y}\right)\dot{\theta}\dot{\psi} + \frac{J_R}{I_x}\dot{\varphi}w_r + \frac{l}{I_y}U_3 \quad (17)$$

$$\ddot{\psi} = \left(\frac{I_x - I_y}{I_z}\right)\dot{\varphi}\dot{\theta} + \frac{1}{I_z}U_4 \quad (18)$$

B. Pysical Parameters of Quadrotor

In this paper, OS4 quadrocopter is used as a quadrotor platform. Table I indicates the physical parameters of this platform. [10]

TABLE I. OS4 PARAMETERS

Parameter	Name	Value
m	Quadrotor mass	0.65 kg
l	Quadrotor arm length	0.23 m
b	Thrust factor	$3.13\text{x}10^{-5}$ Ns^2
d	Drag factor	$7.5\text{x}10^{-7}$ Ns^2
Ix	Inertial moment along x-axis	$7.5\text{x}10^{-3}$ $kg.m^2$
Iy	Inertial moment along y-axis	$7.5\text{x}10^{-3}$ $kg.m^2$
Iz	Inertial moment along z-axis	$1.3\text{x}10^{-2}$ $kg.m^2$
Jr	Moment inertia of propellers	$6.5\text{x}10^{-5}$ $kg.m^2$
w_{max}	Max. rotor speed	1000 rad/sec
tmax	Max torque	0.15 Nm
g	gravity	$9.81 m/s^2$

C. PID Controller Design

In this section, for the OS4 platform, PID control scheme is designed. PID Controller has three parameters. These parameters are proportional, integral and derivative gains. Increasing proportional term Kp decreases the rise time and tends to decrease but could not remove the steady-state error. The integral control term Ki could remove the steady-state error, but it could cause a poor transient response. The derivative control term Kd improves the stability, decreases the overshoot, and leads to a better transient response.

The three crucial flight dynamics parameters are the angles of rotation in three dimensions about the quadrotor's center of gravity (cg), known as pitch, roll and yaw angles We designed various PID controllers for altitude and attitude (roll, pitch and yaw angles). Control method for altitude PID is as below

$$e_z = z_d - z \quad (19)$$

$$U_1 = \frac{m*(g + ezKp + Ki\int e_z dt - Kd\frac{d}{dt}ez)}{\cos\varphi\cos\theta} \quad (20)$$

Fig. 2 shows PID control block diagram for altittude control.

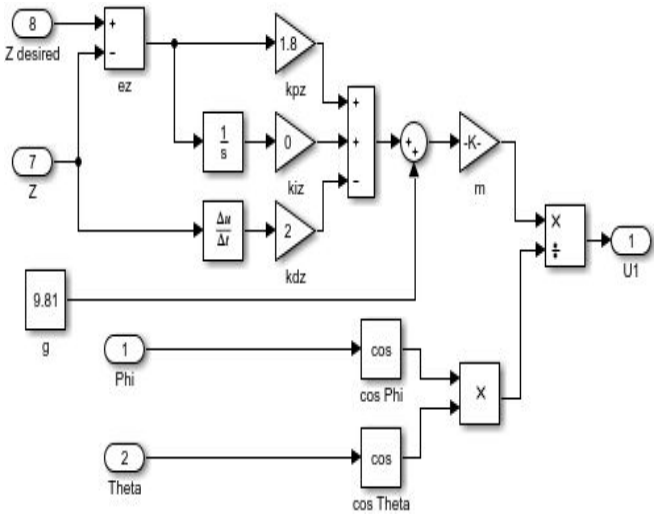


Fig. 2. Block diagram of altitude controller

Table II shows altitude controller parameters.

TABLE II. ALTITUDE CONTROLLER PARAMETERS

Parameter	Value
Kp	1.8
Ki	0
Kd	2

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Control method for roll angle is below.

$$e_\varphi = \varphi_d - \varphi \tag{21}$$

$$U_2 = e_\varphi Kp + Ki \int e_\varphi \, dt - Kd \frac{d}{dt} \varphi) \tag{22}$$

Fig. 3 shows PID control block diagram for roll control. Pitch and yaw control block schemes are also similar except for reference angles.

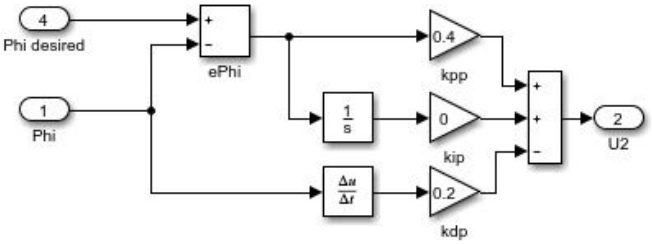


Fig. 3. Block diagram of roll controller

Table III shows PID coefficients for attitude control.

TABLE III. ATTITUDE CONTROLLER PARAMETERS

Parameter	Angle		
	Roll	Pitch	Yaw
Kp	0.4	0.6	0.3
Ki	0	0	0
Kd	0.2	0.2	0.2

e_φ is error in roll angle and it is calculated by taking difference between desired roll angle φ_d and current roll angle φ. Thereafter PID controller is applied to minimize error as indicated in Fig. 3. Kp value is set to reduce the rise time but it cause overshoot in the system. Due to this, Kd is set to decrease overshoot. Ki will only be used for reducing the steady state error when there exist some steady state error in the system. [11]

Control method for pitch angle is below

$$e_\theta = \theta_d - \theta \tag{23}$$

$$U_3 = e_\theta Kp + Ki \int e_\theta \, dt - Kd \frac{d}{dt} \theta) \tag{24}$$

Control method for yaw angle is below

$$e_\varphi = \psi_d - \psi \tag{25}$$

$$U_4 = e_\psi Kp + Ki \int e_\psi \, dt - Kd \frac{d}{dt} \psi) \tag{26}$$

IV. SIMULATION RESULTS

In the previous parts, the mathematical model of quadrotor and PID control method are described. This section contains the simulation results. Fig. 4 shows simulation results for altitude for PID controller.

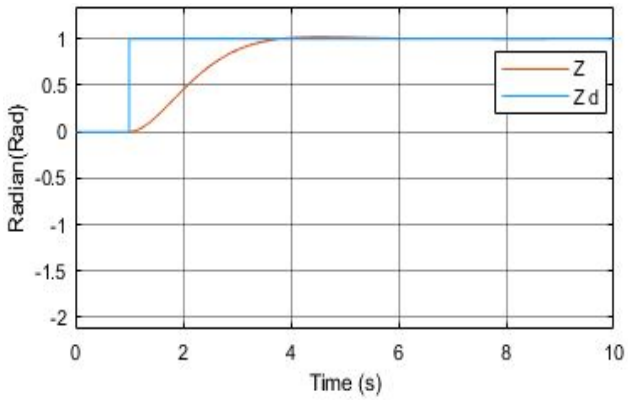


Fig. 4. Simulation results for altitude

Fig. 5 through 7 shows simulation results for roll, pitch and yaw angles which are obtained for PID controller.

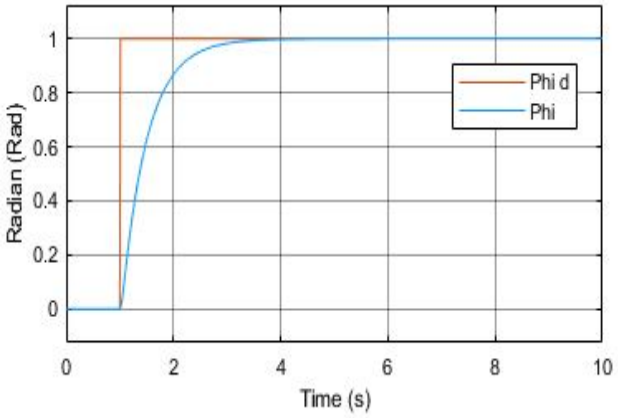


Fig. 5. Simulation results for roll angle

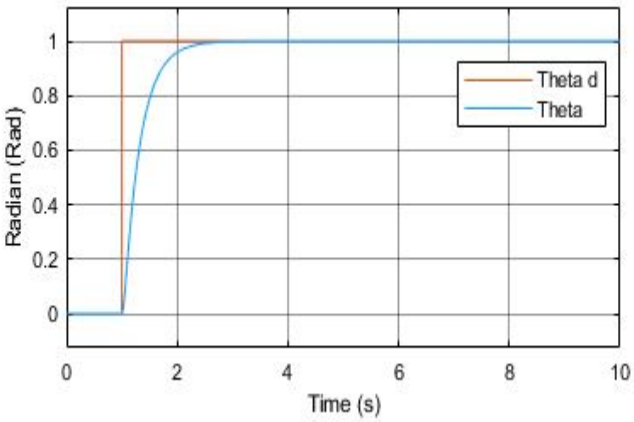


Fig. 6. Simulation results for pitch angle

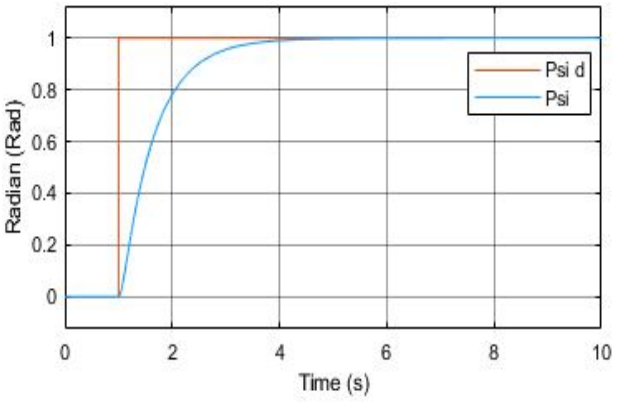


Fig. 7. Simulation results for yaw angle

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Fig. 8 through 11 shows simulation results of each four rotor's torque values.

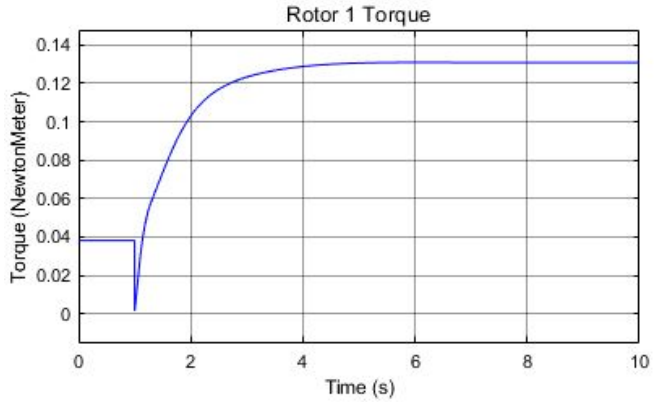


Fig. 8. Rotor 1 Torque

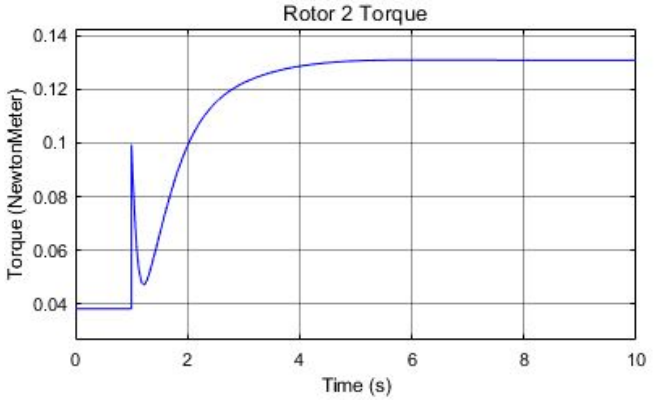


Fig. 9. Rotor 2 Torque

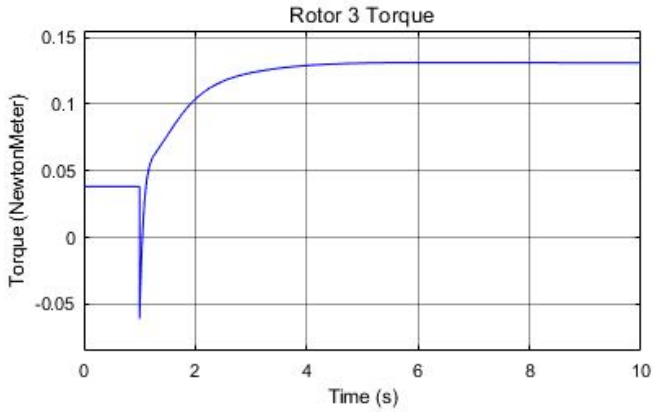


Fig. 10. Rotor 3 Torque

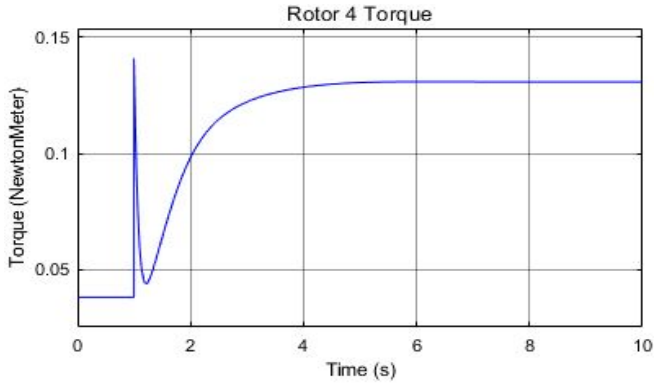


Fig. 11. Rotor 4 Torque

V. Conclusion

In this paper, the nonlinear dynamic model of the quadrotor is formulated. Based on the nonlinear equations, a quadrotor simulator was developed in Simulink. PID contoller is designed for OS4 micro quadrotor platform. Simulation results show that the PID controller provides satisfactory results in tracking the desired reference values. Table I. indicates that each rotor of quadrotor reach 0.15 Nm max. torque value. Torque simulation results prove that rotors' torque values are compatible with quadrotor's physical parameters.

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